

Smoothing and forecasting mortality rates

Iain D Currie¹, Maria Durban² and Paul HC Eilers³

¹Department of Actuarial Mathematics and Statistics, Heriot-Watt University, Edinburgh, UK

²Departamento de Estadística y Econometría, Universidad Carlos III de Madrid, Edificio Torres Quevedo, Leganes, Madrid, Spain

³Department of Medical Statistics, Leiden University Medical Center, Leiden, The Netherlands

Abstract: The prediction of future mortality rates is a problem of fundamental importance for the insurance and pensions industry. We show how the method of *P*-splines can be extended to the smoothing and forecasting of two-dimensional mortality tables. We use a penalized generalized linear model with Poisson errors and show how to construct regression and penalty matrices appropriate for two-dimensional modelling. An important feature of our method is that forecasting is a natural consequence of the smoothing process. We illustrate our methods with two data sets provided by the Continuous Mortality Investigation Bureau, a central body for the collection and processing of UK insurance and pensions data.

Key words: forecasting; mortality; overdispersion; *P*-splines; two dimensions

Data and software link available from: <http://stat.uibk.ac.at/SMIJ>

Received September 2003; revised July 2004; accepted August 2004

1 Introduction

The modelling and projecting of disease incidence and mortality rates is a problem of fundamental importance in epidemiology and population studies generally, and for the insurance and pensions industry in particular. Human mortality has improved substantially over the last century, but this manifest benefit has brought with it additional stress in support systems for the elderly, such as healthcare and pension provision. For the insurance and pensions industry, the pricing and reserving of annuities depends on three things: stock market returns, interest rates and future mortality rates. Likewise, the return from savings for the policyholder depends on the same three factors. In the most obvious way, increasing longevity can only be regarded as a good thing for the policyholder; a less welcome consequence is that annual income from annuities will be reduced. In this article, we consider one of these three factors: the prediction of mortality rates. We have been provided with data sets from two classes of UK insurance business, and we will use these to illustrate our approach to the smoothing and projecting of mortality rates.

Address for correspondence: Iain D Currie, Department of Actuarial Mathematics and Statistics, Heriot-Watt University, Edinburgh EH14 4AS, UK. E-mail: i.d.currie@hw.ac.uk

The requirements of the insurance industry for forecasts of future mortality are daunting indeed, because forecasts up to 50 years ahead are required for pricing and reserving. Human mortality so far ahead depends on the impact of such unknowables as future medical advances, new infectious diseases, and even disasters, both natural and man-made. We make no attempt to take these factors into account and instead simply attempt to forecast future mortality by extrapolating past trends. There are a number of approaches to the problem. One of the oldest methods is based on the forecasting of parameters in some parametric model. For example, the age–period–cohort (APC) model is a well established method of smoothing mortality tables. The classic reference is Clayton and Schifflers (1987). These authors sound warnings about the dangers of overinterpreting the fitted parameters in an APC model and are equally sceptical about the wisdom of extrapolating. Lee and Carter (1992) introduced a simple bilinear model of mortality in which the time dependent component of mortality is reduced to a single index which is then forecast using time series methods. The model is fitted by ordinary least squares (OLS) with the observed log mortality rates as dependent variable. Brouhns *et al.* (2002) improved on the OLS approach by modelling the number of deaths directly by a Poisson distribution and using maximum likelihood for parameter estimation.

Both the APC and the Lee–Carter approach make strong assumptions about the functional form of the mortality surface. There are various fully two-dimensional approaches. Loess (Cleveland and Devlin, 1988) is available directly in the `gam()` function in `Spluse/R` but does not allow extrapolation. De Boor (2001) and Dierckx (1993) construct a two-dimensional regression basis as the Kronecker product of *B*-splines but neither author considers non-normal data or the forecasting problem. Gu and Wahba (1993) and Wood (2003) use thin plate splines, but again forecasting is not available. Our own approach is to use two-dimensional regression splines, specifically *B*-splines with penalties, usually known as *P*-splines (Eilers and Marx, 1996). This approach is used in Eilers and Marx (2003) and Eilers *et al.* (2004) but neither paper considers forecasting.

Eilers and Marx (1996) used *B*-splines as a basis for one-dimensional regression, and we extend this work by using *B*-splines to construct a basis for bivariate regression. This construction gives a basis in two dimensions with local support and hence a fully flexible family of fitted mortality surfaces. The regression approach leads to a generalized linear model (GLM) which is fitted by penalized likelihood. An important feature of this method is that forecasting is a natural consequence of the smoothing process. We consider future values as missing values; the penalization then allows the estimation of future values simultaneously with the fitting of the mortality surface. We will see that the choice of penalty function, which can be of secondary importance in the smoothing of data, is now critical, because it is the penalty function that determines the form of the forecast.

The plan of the article is as follows. In Section 2, we describe our two data sets. In Section 3, we review the *P*-spline approach in one dimension and then show how this extends to two dimensions; the missing value approach to forecasting is also described. In Section 4, we apply our methods to the two data sets; overdispersion is a particular problem with one of the data sets, and we describe one way of dealing with this problem. The article concludes with a critical discussion of our methodology and the implications of the findings for the UK insurance industry.

2 Description of the data

The mortality of insured lives differs from that of the general population and even differs within different classes of business. The Continuous Mortality Investigation Bureau (CMIB) is a central body funded by the UK insurance and pensions industry. UK insurance companies submit claims data to the CMIB for collation and analysis, the principal aim of which is the forecasting of age specific mortality tables for various classes of insured lives. In this article, we consider two of the CMIB data sets, one for male assured lives and one for male pensioners. We describe each of these data sets in turn.

2.1 Male assured lives

For each calendar year (1947–99) and each age (11–100), we have the number of policy claims (deaths) and the number of years lived (the exposure). The claims and exposure data are arranged in matrices Y and E , respectively, whose rows are indexed by age (here 11–100) and whose columns are indexed by year (here 1947–99). One possible difficulty with the data is that claims do not correspond precisely to deaths, because an individual life may hold more than one policy and thus give rise to more than one claim. This is known as the problem of duplicates in insurance terminology and leads to the statistical phenomenon of overdispersion. It is convenient to define $R = Y/E$, the matrix of raw hazards. The problem is to project values of the underlying hazard forward in time.

2.2 Male pensioners

This data set is richer than the previous set because in addition to data Y and E we also have matching data Z and F on amounts. To be precise, the claims in Y give rise to total amounts claimed in Z and the exposures E give rise to total amounts at risk F . These data are available for calendar years 1983–2000 and ages 50–100. We define $R_L = Y/E$, the matrix of raw hazards based on lives, and $R_A = Z/F$, the corresponding quantity based on amounts; here, and in what follows, the suffices L and A indicate lives and amounts, respectively. Clearly, if all policies are for the same amount then $R_L = R_A$. In practice, this is very far from the case, and it is generally found that $R_A < R_L$ (i.e., the mortality of those lives with larger policy amounts is better than those with smaller amounts). The problem of duplicates which is present to a small degree in lives data is an essential feature of amounts data, because the variation between amounts at risk contributes directly to the variation in the total amounts claimed.

3 Smoothing and forecasting mortality tables with P -splines

The method of P -splines is now well established as a method of smoothing in GLMs. Descriptions of the method can be found in the original article of Eilers and Marx (1996) as well as in Marx and Eilers (1998), Eilers and Marx (2002), Currie and

Durban (2002) and elsewhere. The review article by Wand (2003) adopts a mixed model approach and provides a useful bibliography. Theoretical aspects of P -splines are discussed in Wand (1999), Ruppert and Carroll (2000), Aerts *et al.* (2002), Ruppert (2002) and Durban and Currie (2003). Articles which emphasize applications include Marx and Eilers (1999), Parise *et al.* (2001), Coull *et al.* (2001a, 2001b) and Currie and Durban (2002). Wood and Augustin (2002) and Wood (2003) also use penalized regression but with a different basis from that of the present approach.

A succinct summary of the method of P -splines is: 1) use B -splines as the basis for the regression and 2) modify the log likelihood by a difference penalty on the regression coefficients. It is usual to think of smoothing in GLMs in the GAM framework, but we find it helpful to think of the method as a penalized generalized linear model (PGLM). There are two reasons for this: model building and computation. In a PGLM, we define a regression matrix, an error distribution, a link function (GLM) and a penalty matrix; computationally our method extends the GLM scoring algorithm but retains the iterative weighted least squares (IWLS) form. Eilers and Marx (1996) showed how to smooth Poisson data in one dimension. We extend their method to two dimensions by constructing regression and penalty matrices appropriate for two-dimensional modelling. This section discusses the basic one-dimensional model and the choice of the various parameters implicit in smoothing with P -splines, the definition of the two-dimensional regression matrix, the definition of the penalty matrix and finally, the method of extrapolation.

3.1 Smoothing in one dimension

For the benefit of readers unfamiliar with smoothing using P -splines, we present a short introduction to the method. Suppose we have data (y_i, e_i, x_i) , $i = 1, \dots, n$, on a set of lives all aged 65, say, where y_i is the number of deaths in year x_i and e_i is the exposed to risk. Let $\mathbf{y}' = (y_1, \dots, y_n)$, $\mathbf{e}' = (e_1, \dots, e_n)$, and so on. We suppose that the number of deaths y_i is a realization of a Poisson distribution with mean $\mu_i = e_i \theta_i$. For an introduction to P -splines with normal errors refer to Eilers and Marx (1996) or Marx and Eilers (1998). The left hand panel of Figure 1 shows a plot of the logarithm of the raw forces of mortality, $\hat{\theta}_i = y_i/e_i$. We seek a smooth estimate of $\boldsymbol{\theta} = (\theta_i)$. A classical approach might be to fit a GLM with quadratic regression term that is, $\log \boldsymbol{\mu} = \log \mathbf{e} + \log \boldsymbol{\theta} = \log \mathbf{e} + \mathbf{X} \mathbf{a}$, where $\log \mathbf{e}$ is an offset in the regression term; the result is shown in the left hand panel. This simple model uses $\{1, x, x^2\}$ as basis functions. A more flexible basis is provided by a set of cubic B -splines $\{B_1(x), \dots, B_K(x)\}$ and such a basis is shown for $K = 8$ in the right panel of Figure 1; each B -spline consists of cubic polynomial pieces smoothly bolted together at points known as knots. We are still in the framework of classical regression with regression matrix, $\mathbf{B}$. The rows of $\mathbf{B}$ are the values of B -splines in the basis evaluated at each year in turn; thus, for example, the estimate of $\log \theta$ at 1970 is

$$0.0817 \times \hat{a}_3 + 0.6267 \times \hat{a}_4 + 0.2901 \times \hat{a}_5 + 0.0016 \times \hat{a}_6 \quad (3.1)$$

as can be seen from the dashed line in the right panel of Figure 1. In general $B_1 = 1$, $B \geq 0$ and $\mathbf{B}$ is a banded matrix, so the fitted values $\mathbf{B}\hat{\mathbf{a}}$ are weighted averages

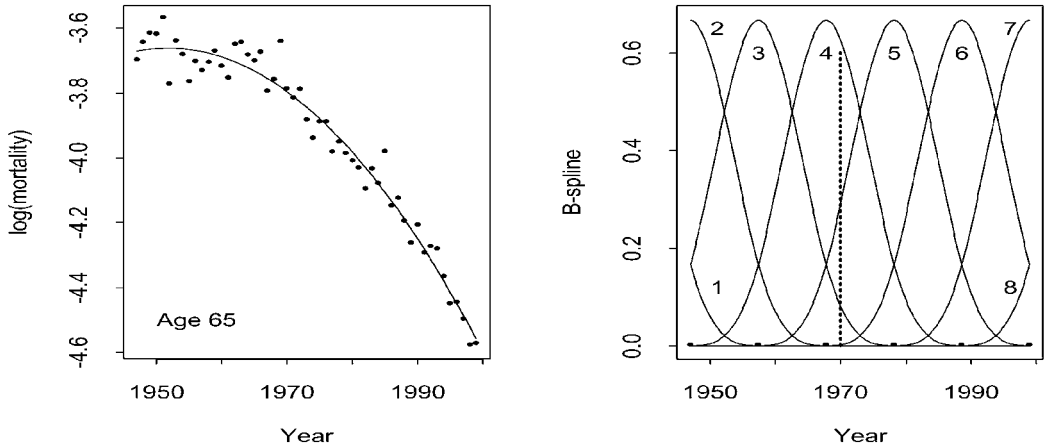


Figure 1 Left panel: observed log(mortality) and fitted quadratic linear predictor; right panel: a basis of $K = 8$ cubic B-splines with knots

of local subsets of the coefficients; interestingly enough, the same weights applied to the knot positions $\mathbf{k}$ recover the $\mathbf{x}$ values because $\mathbf{B}\mathbf{k} = \mathbf{x}$.

The left panel of Figure 2 shows the result of fitting a GLM with $K = 23$ B-splines in the basis. Evidently, the data have been undersmoothed. The plot also shows the regression coefficients ($\hat{a}_2, \dots, \hat{a}_{22}$) plotted at the maximum value of their corresponding B-spline ($\hat{a}_1 = -3.05$ and $\hat{a}_{23} = -2.45$ are omitted because their extreme values distort the scale of the plot). Eilers and Marx (1996) observed that the undersmoothing in the

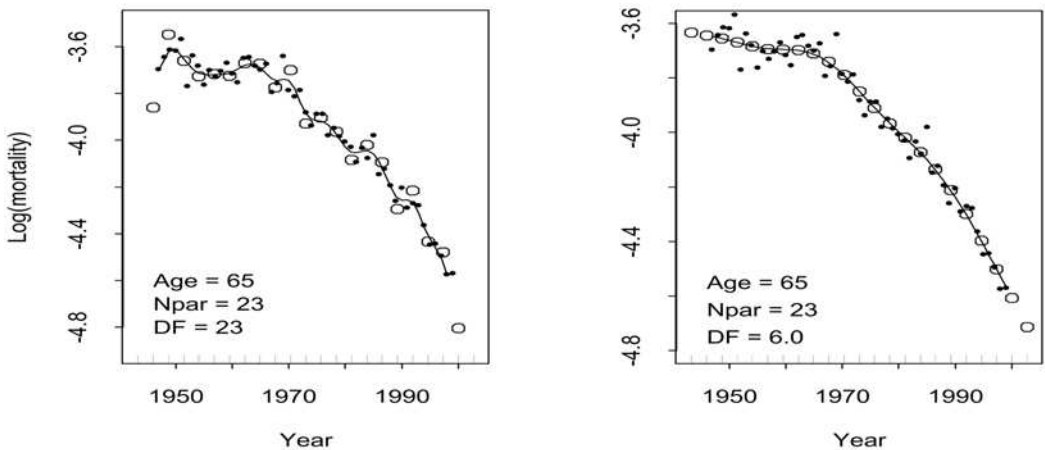


Figure 2 Left panel: observed log(mortality), fitted regression with a basis of $K = 23$ cubic B-splines and regression coefficients, $\circ$; right panel: as left panel but with P-spline regression (knot positions are shown by vertical tick marks)

left panel of Figure 2 was a result of the erratic behaviour of the $\hat{a}_k$ and they proposed penalizing this behaviour by placing a difference penalty on adjacent a_k , as in

$$(a_1 - 2a_2 + a_3)^2 + \cdots + (a_{K-2} - 2a_{K-1} + a_K)^2 = \mathbf{a}'\mathbf{D}'\mathbf{D}\mathbf{a} \quad (3.2)$$

where $\mathbf{D}$ is a difference matrix of order 2; this defines a quadratic penalty, but linear and cubic penalty functions are also possible. The penalty function is incorporated into the log likelihood function to give the penalized log likelihood

$$\ell_p = \ell(\mathbf{a}; \mathbf{y}) - \frac{1}{2} \mathbf{a}'\mathbf{P}\mathbf{a} \quad (3.3)$$

$$\ell_p = \ell(\mathbf{a}; \mathbf{y}) - \frac{1}{2} \lambda \mathbf{a}'\mathbf{D}'\mathbf{D}\mathbf{a} \quad (3.4)$$

where $\ell(\mathbf{a}; \mathbf{y})$ is the usual log likelihood for a GLM, $\mathbf{P} = \lambda \mathbf{D}'\mathbf{D}$ is the penalty matrix and λ is the smoothing parameter.

Maximizing Equation (3.3) gives the penalized likelihood equations

$$\mathbf{B}'(\mathbf{y} - \boldsymbol{\mu}) = \mathbf{P}\mathbf{a} \quad (3.5)$$

which can be solved with the penalized version of the scoring algorithm

$$(\mathbf{B}'\tilde{\mathbf{W}}\mathbf{B} + \mathbf{P})\hat{\mathbf{a}} = \mathbf{B}'\tilde{\mathbf{W}}\mathbf{B}\tilde{\mathbf{a}} + \mathbf{B}'(\mathbf{y} - \tilde{\boldsymbol{\mu}}) \quad (3.6)$$

where $\mathbf{B}$ is the regression matrix, $\mathbf{P}$ is the penalty matrix, $\tilde{\mathbf{a}}$, $\tilde{\boldsymbol{\mu}}$ and $\tilde{\mathbf{W}}$, the diagonal matrix of weights, denote current estimates and $\hat{\mathbf{a}}$ denotes the updated estimate of $\mathbf{a}$. Notice that Equations (3.5) and (3.6) are the standard likelihood equations and scoring algorithm, respectively, for a GLM adjusted for the quadratic term in Equation (3.3). Note that Equation (3.6) corrects a misprint in Equation (16) in Eilers and Marx (1996). In the case of Poisson errors, we have $\tilde{\mathbf{W}} = \text{diag}(\tilde{\boldsymbol{\mu}})$. The algorithm (3.6) can also be written as

$$(\mathbf{B}'\tilde{\mathbf{W}}\mathbf{B} + \mathbf{P})\hat{\mathbf{a}} = \mathbf{B}'\tilde{\mathbf{W}}\tilde{\mathbf{z}} \quad (3.7)$$

where $\tilde{\mathbf{z}} = \tilde{\boldsymbol{\eta}} + \tilde{\mathbf{W}}^{-1}(\mathbf{y} - \tilde{\boldsymbol{\mu}})$ which emphasizes the connection between this algorithm and the standard IWLS algorithm for a GLM.

It follows from Equation (3.7) that the hat-matrix is given by

$$\mathbf{H} = \mathbf{B}(\mathbf{B}'\hat{\mathbf{W}}\mathbf{B} + \mathbf{P})^{-1}\mathbf{B}'\hat{\mathbf{W}} \quad (3.8)$$

and the trace of the hat-matrix, $\text{tr}(\mathbf{H})$, is a measure of the effective dimension or degrees of freedom of the model (Hastie and Tibshirani, 1990: 52). The approximate variance of $\mathbf{B}\hat{\mathbf{a}}$ is given by

$$\text{Var}(\mathbf{B}\hat{\mathbf{a}}) \approx \mathbf{B}(\mathbf{B}'\hat{\mathbf{W}}\mathbf{B} + \mathbf{P})^{-1}\mathbf{B}' \quad (3.9)$$

One justification of Equation (3.9) is that the P -spline estimator of $\mathbf{Ba}$ can be derived from a Bayesian perspective and Equation (3.9) coincides with its Bayesian variance (Lin and Zhang, 1999).

The user of P -splines has a number of choices to make: 1) the number of knots, the degree of the P -spline and the order of the penalty and 2) the smoothing parameter. The parameters in 1) we call the P -spline parameters and denote them by ndx , $bdeg$ and $pord$. We will see that when forecasting with P -splines, the forecast values depend critically on the order of the penalty. The choice of the other P -spline parameters, ndx and $bdeg$, can be less critical because different choices often lead to similar fitted smooth functions. Eilers and Marx (1996), Ruppert (2002) and Currie and Durban (2002) discuss the choice of the P -spline parameters; the following simple rule-of-thumb is often sufficient: with equally spaced data, use one knot for every four or five observations up to a maximum of 40 knots to guide the choice of the number of knots parameter, ndx , (Strictly $ndx - 1$ is the number of internal knots in the domain of x .); use cubic splines ($bdeg = 3$) and a quadratic penalty ($pord = 2$).

We will use the Bayesian information criterion, BIC, (Schwarz, 1978) to choose the smoothing parameter where

$$\text{BIC} = 2 \text{Dev} + \log n \text{Tr} \quad (3.10)$$

here, Dev is the deviance in a GLM, $\text{Tr} = \text{tr}(\mathbf{H})$ is the effective dimension of the fitted model, as in Equation (3.8) and n is the number of observations. Other possibilities are Akaike information criterion (AIC) or generalised cross-validation (GCV), but there is evidence that AIC and GCV tend to undersmooth the data (Hurvich *et al.*, 1998; Lee, 2003). BIC penalizes model complexity more heavily than AIC particularly, as in our examples, when n is large. (Refer Chatfield (2003) for a useful discussion on criteria for model choice). An alternative approach is to express the PGLM as a generalized linear mixed model (GLMM) and this also leads to an estimate of the smoothing parameter.

The right panel of Figure 2 shows the result of fitting the PGLM. The P -spline parameters were: $bdeg = 3$ (cubic B -splines), $pord = 2$ (second order penalty) and $ndx = 20$, which gives $K = ndx + bdeg = 23$ B -splines in the basis. The optimal value of the smoothing parameter, $\hat{\lambda} = 3900$, was chosen via BIC. This level of smoothing has the effect of reducing the degrees of freedom from 23 (the number of fitted parameters) to about 6.

We close this introduction with a few general comments about the method of P -splines. The method has a number of attractive features. First, the method is an example of a low rank smoother, because the order of the system of Equations (3.6) is equal to the number of B -splines in the basis, and this is generally much less than the number, n , of data points (smoothing splines use matrices of order n). Secondly, the algorithm (3.6) or (3.7) is computationally efficient, because it is essentially the same as the IWLS algorithm used to fit a GLM. Thirdly, the regression form for the mean ($\log \mu = \log e + \mathbf{Ba}$) makes regression-style model building straightforward (Section 4.2).

3.2 Two-dimensional regression matrix

One of the principal motivations behind the use of B -splines as the basis of regression is that it does not suffer from the lack of stability that can so bedevil ordinary polynomial regression. The essential difference is that B -splines have local nonzero support in contrast to the polynomial basis for standard regression. We seek to construct a basis for two-dimensional regression with local support analogous to the way that B -splines provide a basis for one-dimensional regression.

Suppose we have a two-dimensional regression problem with regression variables x_1 and x_2 where the data Y are available over an $m \times n$ grid; the rows of Y are indexed by x_1 and the columns by x_2 (age and year, respectively, in our example). For the purpose of regression, we suppose that the data are arranged in column order, that is, $y = \text{vec}(Y)$. Consider a simple polynomial model with linear predictor defined by $1 + x_1 + x_2 + x_2^2 + x_1x_2 + x_1x_2^2$. Let X be the regression matrix for this model. Now consider two one-dimensional models, one in x_1 for a column y_c of Y and one in x_2 for a row y_r of Y . Let X_1 be the regression matrix corresponding to the model with linear predictor $1 + x_1$ and, in a similar fashion, let X_2 be the regression matrix corresponding to the model with linear predictor $1 + x_2 + x_2^2$. We observe that the model formula $1 + x_1 + x_2 + x_2^2 + x_1x_2 + x_1x_2^2$ factorizes into $(1 + x_2 + x_2^2)(1 + x_1)$ and it follows that

$$X = X_2 \otimes X_1 \quad (3.11)$$

as may be easily checked directly. Here $\otimes$ denotes the Kronecker product of two matrices (Searle, 1982; 265).

We refer to a model for a column or row of Y as a marginal model. Thus Equation (3.11) says that the regression matrix for a polynomial regression in x_1 and x_2 , whose model formula can be written as the product of two marginal models, can be written as the Kronecker product of the regression matrices of the marginal models. In ordinary two-dimensional regression, this observation seems little more than a curiosity, but we use exactly this idea to produce a two-dimensional regression matrix with local support. Let $B_a = B(x_a)$, $n_a \times c_a$, be a regression matrix of B -splines based on the explanatory variable for age x_a and similarly, let $B_y = B(x_y)$, $n_y \times c_y$, be a regression matrix of B -splines based on the explanatory variable for year x_y . The regression matrix for our two-dimensional model is the Kronecker product

$$B = B_y \otimes B_a \quad (3.12)$$

This formulation assumes that the vector of observed claim numbers $y = \text{vec}(Y)$, (this corresponds to how Splus stores a matrix). In the case of assured lives, the data are on a 90×53 grid, and the matrices B_a and B_y are typically 90×20 and 53×10 when B will be 4770×200 .

Figure 3 gives a feel for what a Kronecker product basis looks like. The age-year grid is populated by a set of overlapping hills which are placed at regular intervals over the region. Each hill is the Kronecker product of two one-dimensional hills (B -splines), one in age and one in year. For clarity, only a subset of hills from a small basis is shown in

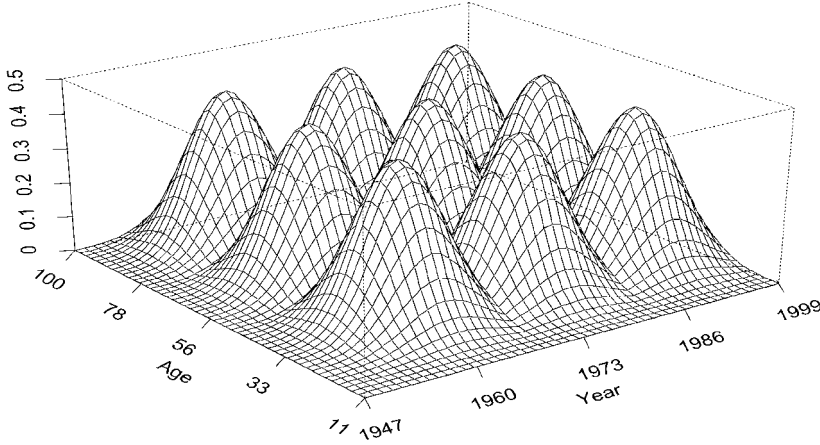


Figure 3 Two-dimensional Kronecker product cubic B -spline basis

Figure 3, but in practice there are about 200 such hills which give a dense covering of the age–year region, and this results in a flexible basis for two-dimensional regression.

3.3 Two-dimensional penalty matrix

The regression matrix $B = B_y \otimes B_a$ defined in Equation (3.12) has an associated vector of regression coefficients $\mathbf{a}$ of length $c_a c_y$. We arrange the elements of $\mathbf{a}$ in $c_a \times c_y$ matrix A , where $\mathbf{a} = \text{vec}(A)$ and the columns and rows of A are given by

$$A = (\mathbf{a}_1, \dots, \mathbf{a}_{c_y}) \quad A' = (\mathbf{a}'_1, \dots, \mathbf{a}'_{c_a}) \quad (3.13)$$

It follows from the definition of the Kronecker product that the linear predictor corresponding to the j th column of Y can be written

$$\sum_{k=1}^{c_y} b_{jk}^y B_a \mathbf{a}_k \quad (3.14)$$

where $B_y = (b_{ij}^y)$. In other words, the linear predictors of the columns of Y can be written as linear combinations of c_y smooths in age. This suggests that we should apply a roughness penalty to each of the c_y columns of A . This gives the left hand side of

$$\sum_{j=1}^{c_y} \mathbf{a}'_j D'_a D_a \mathbf{a}_j = \mathbf{a}' (I_{c_y} \otimes D'_a D_a) \mathbf{a} \quad (3.15)$$

as an appropriate penalty, where D_a is a difference matrix acting on the columns of A ; the right hand side follows immediately from the definition of the Kronecker product and is a convenient form in terms of the original regression vector $\mathbf{a}$. In a similar

fashion, by considering the linear predictor corresponding to the i th row of Y , we can show that the corresponding penalty on the rows of A can be written

$$\sum_{i=1}^{c_a} \mathbf{a}_i' D_y' D_y \mathbf{a}_i = \mathbf{a}' (D_y' D_y \otimes I_{c_a}) \mathbf{a} \quad (3.16)$$

where D_y is a difference matrix acting on the rows of A . The regression coefficients $\mathbf{a}$ are estimated by maximizing the penalized log likelihood (3.3) where B is given by Equation (3.12) and the penalty matrix P by

$$P = \lambda_a I_{c_y} \otimes D_a' D_a + \lambda_y D_y' D_y \otimes I_{c_a} \quad (3.17)$$

λ_a and λ_y are the smoothing parameters in age and year, respectively.

3.4 Forecasting with P -splines

We treat the forecasting of future values as a missing value problem and estimate the fitted and forecast values simultaneously. In one dimension we have data $\mathbf{y}_1$ and $\mathbf{e}_1$ for n_1 years for some age. Let B_1 be the B -spline regression matrix in a P -spline model of mortality. Suppose that we wish to forecast n_2 years into the future. We extend the set of knots used to compute B_1 and compute the regression matrix B for $n_1 + n_2$ years. Then B has the following form

$$B = \begin{bmatrix} B_1 & 0 \\ B_2 & B_3 \end{bmatrix} \quad (3.18)$$

Let $\mathbf{y}_2$ and $\mathbf{e}_2$ be arbitrary future values and let $\mathbf{y}' = (\mathbf{y}_1', \mathbf{y}_2')$ and $\mathbf{e}' = (\mathbf{e}_1', \mathbf{e}_2')$ hold the observed and missing data. We estimate the regression coefficients $\mathbf{a}$ with Equation (3.3) computed over the observed data only, that is,

$$\ell_p = \ell(\mathbf{a}; \mathbf{y}_1) - \frac{1}{2} \mathbf{a}' P \mathbf{a} \quad (3.19)$$

If we define a weight matrix $V = \text{blockdiag}(I, 0)$ where I is an identity matrix of size n_1 and 0 is a square matrix of 0s of size n_2 then the form (3.18) of the regression matrix B enables the penalized scoring algorithm (3.6) to be written

$$(B' V \tilde{W} B + P) \hat{\mathbf{a}} = B' V \tilde{W} B \tilde{\mathbf{a}} + B' V (\mathbf{y} - \tilde{\boldsymbol{\mu}}) \quad (3.20)$$

a convenient form for calculation because fitting and forecasting can be done simultaneously.

The form of the penalized log likelihood (3.19) emphasizes that it is the penalty function which allows the forecasting process to take place and that the form of the penalty determines the form of the forecast. The penalty on the coefficients $\mathbf{a}$ ensures smoothness of the coefficients, and smooth forecast values result. We use the age 65

data to illustrate the procedure. We have data from 1947 to 1999 and forecast to 2050. We make two comments on the resulting Figure 4: first, the order of the penalty has no discernible effect on the regression coefficients in the region of the data; second, the order of the penalty has a dramatic effect on the extrapolated values. The method works by extrapolating the regression coefficients, and these extrapolations are constant, linear or quadratic depending on the order of the penalty.

We mention some invariance properties of the forecasting procedure in one dimension. The following quantities do not change whether we include or exclude the missing values to be forecast: the estimates of the regression coefficients and the fitted values within the range of the training set, the trace of the fitted model and the optimal value of the smoothing parameter. These results can be proved using the partition form (3.18) of the regression matrix. Thus, estimation and prediction can be divided into a two stage procedure, that is, fitting to the training set and then forecasting the missing values; we prefer the convenience of Equation (3.20).

In two dimensions, the penalty function again enables the forecast to take place, and fitting and forecasting can again be done with Equation (3.20). However, the forecast coefficients are only approximately constant, linear or quadratic in the year direction (as determined by the order of the penalty) because the age penalty tends to maintain the age structure across ages. Thus, the choice of $pord$ has major implications for the extrapolation. Our strong preference is for $pord = 2$: $pord = 1$ does not sit comfortably

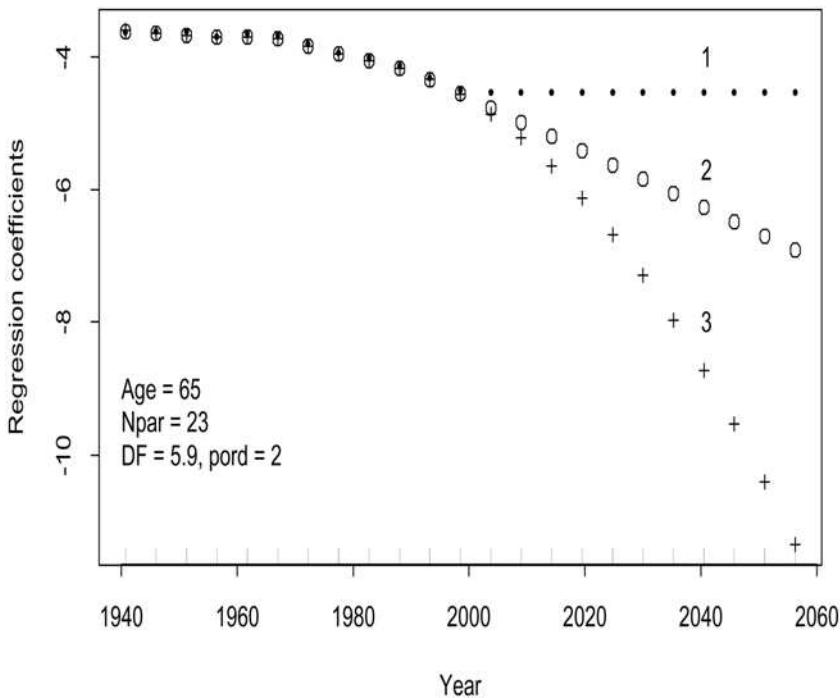


Figure 4 Fitted regression coefficients $\bullet$, $\circ$ and $+$ for $pord = 1, 2, 3$, respectively

with Figure 2 and we are uneasy about using a quadratic extrapolation over such a long time period. We will use $pord = 2$ when we consider the extrapolation of mortality tables in the next section.

4 Applications

The previous section set out the theory for extrapolating with P -splines and we used the age 65 data to illustrate the technique in one dimension. However, the main purpose of this article is the extrapolation of mortality tables, and we now apply our method to the two data sets described in Section 2. There is overdispersion in both data sets. The overdispersion is slight in the assured lives data set, and to simplify the presentation, we ignore it in Section 4.1. However, the overdispersion in the pensioner data cannot be ignored, and we build a joint model for the mean and dispersion of the log hazard in Section 4.2. Finally in Section 4.2, we use the regression form of our model to build a consistent joint model for the future mortality as determined by both lives and amounts.

4.1 Male assured lives data

We start with a validation exercise and use the 1947–74 data to predict the 1975–99 rates. We set $bdeg = 3$, cubic B -splines, and $pord = 2$, second order penalties, and then selected the values of ndx by minimizing BIC over the 3×3 grid with each ndx taking the values 5, 10 and 20 in turn; ndx was set to 10 for age and 10 for year. This choice of ndx gave a model with 169 fitted parameters which was reduced to an effective dimension of 41.2 after smoothing parameter selection with BIC. Figure 5 gives a general impression of the mortality surface, while Figure 6 shows cross-sections

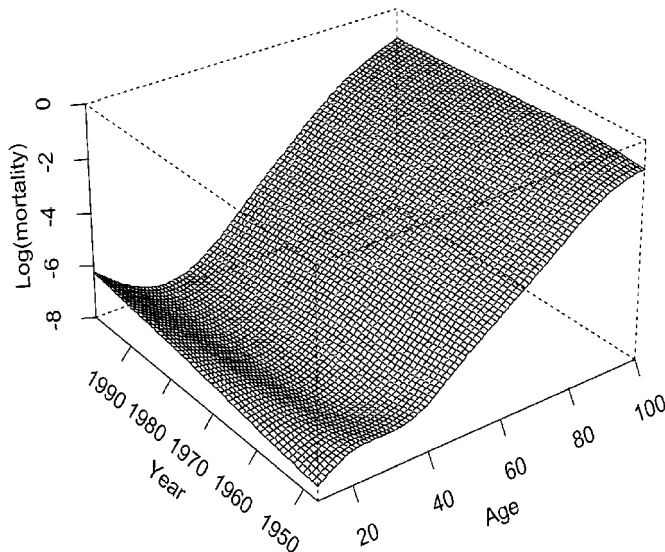


Figure 5 Fitted and extrapolated log(mortality) surface

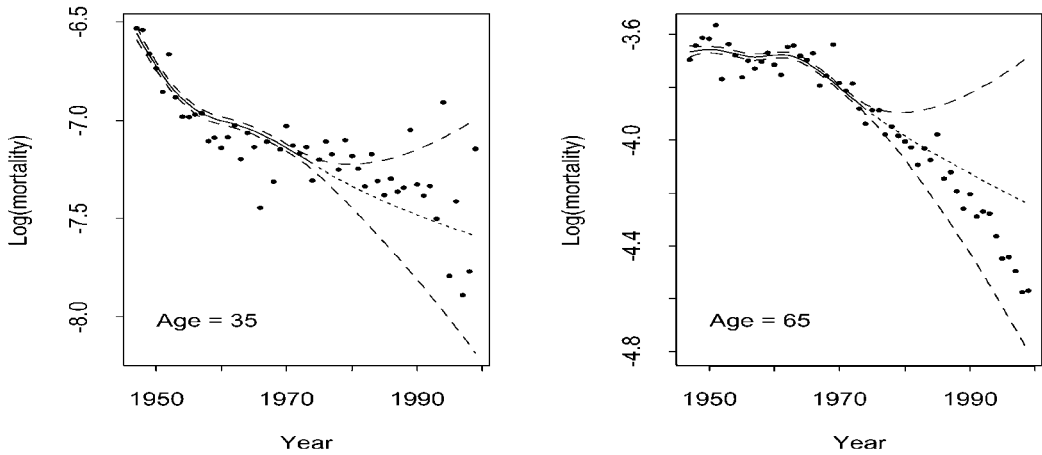


Figure 6 Fitted and extrapolated log(mortality) with 95% confidence intervals. Left panel: age 35, right panel: age 65

(ages 35 and 65) of the surface. It is clear from both figures that the fitted surface is highly nonadditive and so a proper description of the mortality surface requires a flexible two-dimensional model. Confidence intervals for the fitted and forecast values are computed simultaneously from Equation (3.9) and are included in Figure 6.

We make two comments on Figure 6. First, even with the benefit of hindsight, it is hard to see how the sharp falls in mortality that occurred from the 1970s to the present could have been predicted back in the 1970s. Second, although our extrapolated rates are generally too high for ages over 50, the observed rates do lie comfortably within the 95% confidence intervals. We will comment further on these points in the final section of the article.

We turn now to the industry requirement for extrapolations up to the year 2050. The values of ndx were set to 10 for age and 20 for year (comparable to ndx values of 10 in the validation exercise), which gave 299 parameters to model the mortality surface; smoothing parameter selection reduced this to an effective dimension of 64. The left panel of Figure 7 shows fitted and extrapolated values from 30 to 90 at 10 year age intervals. Again, the nonadditive nature of the fitted surface is apparent. One obvious point of comment is the crossing of the age 30 and 40 mortalities around 2035. The data support this, because mortality around age 30 has generally flattened out over the last 20 years, whereas that around 40 has fallen steadily. Of course, it is not unknown for mortality at younger ages to be heavier than that at older ages; the 'accident bump' for males aged about 20 is caused by identifiable behaviour at this age. One surmises that lifestyle choices too have halted the improvement in mortality of those aged about 30. However, if one does not accept that the mortalities at age 30 and 40 will cross over at some future date then an ad hoc fix could be implemented; for example, one possibility is to use confidence intervals. Mortality for ages 60–70 is of particular interest to the insurance industry because many policies mature at these ages. The right hand panel of Figure 7 shows fitted and extrapolated values, and we see mortality increasing in a smooth nonadditive fashion from ages 60 to 70.

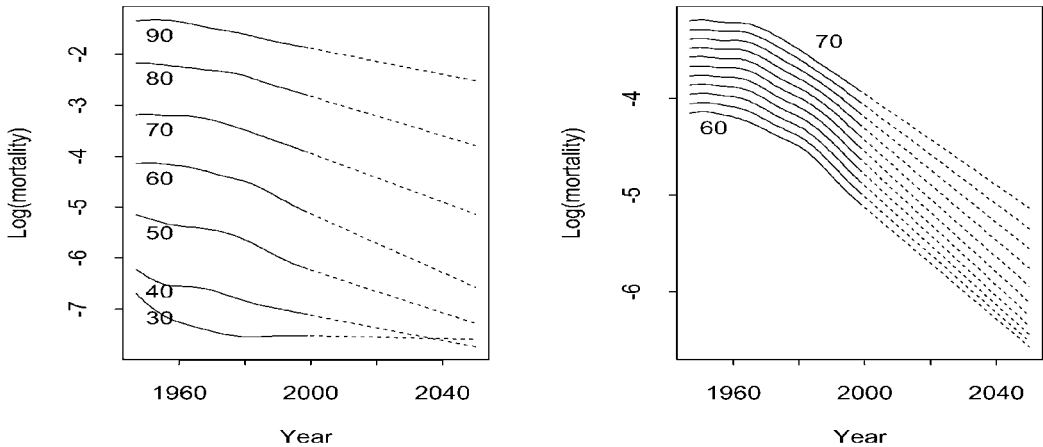


Figure 7 Fitted (solid line) and extrapolated (dashed line) log(mortality). Left panel: ages 30–90 in 10 year bands; right panel: ages 60–70

We present a comparison between the P -spline and the Lee–Carter approaches to predicting future mortality. The Lee–Carter model assumes that

$$Y_{ij} \sim \mathcal{P}(E_{ij}\theta_{ij}) \quad \log \theta_{ij} = \alpha_i + \beta_i \kappa_j \quad \sum \kappa_j = 0 \quad \sum \beta_i = 1 \quad (4.1)$$

where the constraints ensure that the model is identifiable. The model specifies that the log of the force of mortality is a bilinear function of age, i , and time, j . The parameters have simple interpretations: α represents overall mortality, κ represents the time trend and β is an age specific modification to the time effects. We follow Brouhns *et al.* (2002) and fit the model by maximum likelihood. Forecasting is achieved by assuming that the age terms α and β remain fixed at their estimated values and forecasting κ with an ARIMA time series.

Figure 8 compares fits and forecasts for four ages. The difference in the two sets of forecasts is striking with the Lee–Carter method predicting much heavier falls in mortality than P -splines. The Lee–Carter forecasts fall within the 95% confidence funnels for the P -spline forecasts and so in this sense, the two forecasts are consistent with each other. In terms of fit to the data, the P -spline model is superior with a GLM deviance of 8233 compared to 9203 for the Lee–Carter model. The P -spline model is a local two-dimensional method, so the fitted mortality surface responds to local changes in the observed mortality rates; this explains the lower deviance of the P -spline model. We also note that the lower deviance is achieved with a smaller number of parameters: the P -spline model has an effective dimension of about 64, while the Lee–Carter model has $2 \times 90 + 53 - 2 = 231$ parameters.

4.2 Male pensioner data

The male pensioner data consist of two matching data sets, one for lives and one for amounts. The lives data consist of data matrices Y on claim numbers and E on exposures, as in the case of the assured lives data just discussed. In addition, we have data matrices Z and F which give the total amount claimed and the total amount at risk.

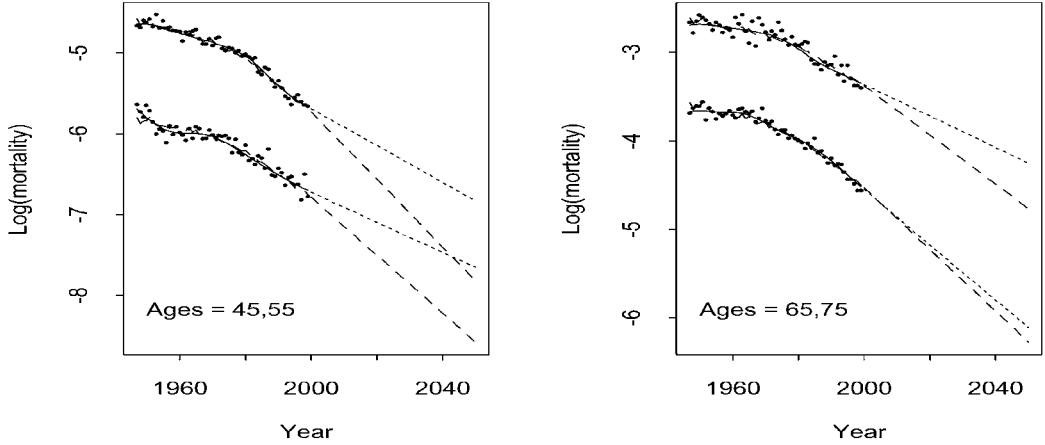


Figure 8 Observed, fitted and projected log(mortality): *P*-spline as ---, Lee-Carter as ----

The rows of these matrices are indexed by age, 50–100, and the columns by year, 1983–2000. One possibility is to model Z/F as lognormal or gamma though the zeros in Z at all ages under 65 make for some difficulties; further, this approach ignores the information in the lives data set. Our plan is to model the amounts data as an overdispersed Poisson distribution. We use a two stage procedure.

Stage 1. We define $A = F/E$, the mean amount at risk per life. Then the matrix of raw hazards by amounts R_A can be written

$$R_A = \frac{Z}{F} = \frac{(Z/A)}{(F/A)} = \frac{Z^*}{E} \quad (4.2)$$

where $Z^* = Z/A$. If all policies are for the same amount, then $Z^* = Y$ and so our first approximation is to assume $Z_{ij}^* \sim \mathcal{P}(E_{ij}\theta_{ij})$ and smooth the mortality table exactly as in Section 4.1. Let $\bar{Z}^*$ denote the smoothed values of Z^* , that is, $\bar{Z}^*$ is an estimate of the mean of Z^* .

Stage 2. The distribution of Z^* has larger variance than that of Y , because all policies are not for the same amount. Furthermore, it is known that the sums assured depend strongly on age so we hope to improve on the assumption in stage 1 by assuming that

$$\frac{Z_{ij}^*}{\phi_i} \sim \mathcal{P}\left(\frac{E_{ij}}{\phi_i}\theta_{ij}\right) \quad (4.3)$$

where ϕ_i is an age dependent overdispersion parameter. We use the medians of the rows of $(Z^* - \bar{Z}^*)^2 / \bar{Z}^*$ as an initial estimate of the ϕ_i ; medians are preferred to means for robustness reasons. The left panel of Figure 9 gives a plot of these medians together with their smoothed values obtained by a one-dimensional *P*-spline smooth on the log scale; we take these smoothed values as our estimate of the overdispersion

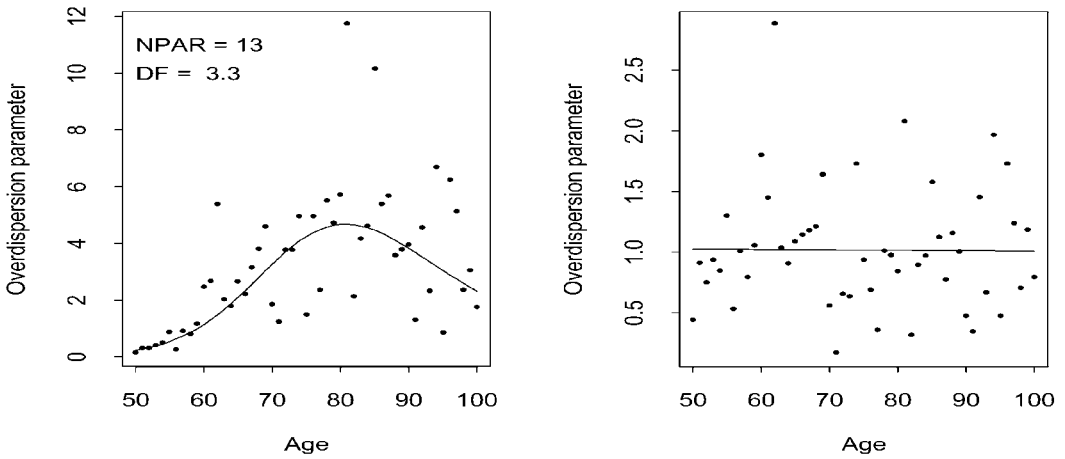


Figure 9 Overdispersion parameters by age. Left panel: initial estimates of ϕ_i with P -spline smooth; right panel: estimates of ϕ_i after scaling as in Equation (4.3)

parameters ϕ_i in Equation (4.3). One further iteration of Equation (4.3) gives the right hand panel of Figure 9; the overdispersion has been removed.

The effect of the scaling in both Equation (4.2) and Equation (4.3) is to reduce the exposure from the total amounts at risk in F to an effective sample size approximated initially by F/A . One consequence in this ‘reduction in sample size’ is that the resulting fit is considerably stiffer: the degrees of freedom for the three fits referred to in the preceding paragraphs are 42 (stage 1), 19 and 17 (stage 2).

We also used the method of Section 4.1 to forecast the mortality using the lives data, and Figure 10 gives a comparison for two ages of the log(mortality) using the two

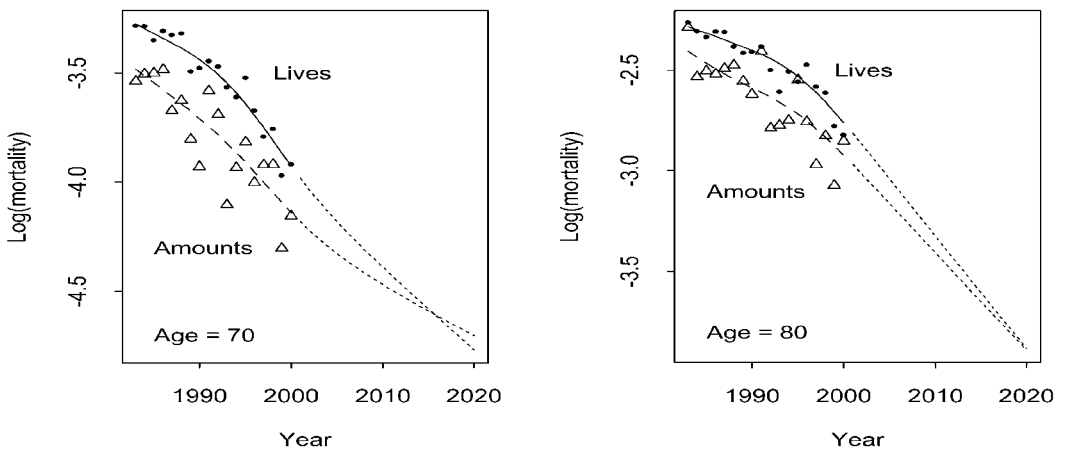


Figure 10 Observed, smoothed and extrapolated mortality (on the log scale) separately by lives and amounts: ages 70 and 80

data sets. We make three comments: first, the mortality by lives is heavier than by amounts, in line with industry experience; secondly, the overdispersion in the amounts data is evident from both plots; thirdly, the predictions both cross over. This last comment may give cause for concern. Crossing arises simply because in the 1990s, the improvement in mortality by lives is faster than that by amounts; crossing results. However, it is an accepted fact in the industry that mortality by lives is and will be heavier than that by amounts. This suggests that we build a joint model for mortality by lives and by amounts, and the regression form of our model makes this particularly simple. For example, we can fit surfaces with parallel cross sections in age (i.e., we force the fitted curves in Figure 10 to be parallel). We take lives as the baseline and use $\eta_L = B_y \otimes B_a a$ as linear predictor. For amounts, we add an age dependent constant (the gap between the mortality curves in Figure 10). If we smooth these gaps then we get $\eta_A = B_y \otimes B_a a + 1_y \otimes B_a g$ as linear predictor where 1 is a vector of 1s, that is, $\eta_A = \eta_L + 1_y \otimes B_a g$. Now we have three smoothing parameters to choose: the mortality surface (determined by a) is subject to a penalty of the form (3.17) while the gap (determined by g) is subject to the basic penalty function in Equation (3.3). The explicit regression formulation is

$$y = \begin{bmatrix} y_L \\ y_A \end{bmatrix} \quad e = \begin{bmatrix} e_L \\ e_A \end{bmatrix} \quad B = \begin{bmatrix} B_y \otimes B_a & 0 \\ B_y \otimes B_a & 1_y \otimes B_a \end{bmatrix} \quad P = \begin{bmatrix} P_1 & 0 \\ 0 & P_2 \end{bmatrix} \quad (4.4)$$

where P_1 is given by Equation (3.17) and P_2 by Equation (3.3); the claim numbers and exposures are given by $y_L = \text{vec}(Y)$ and $e_L = \text{vec}(E)$ and the claim amounts, y_A , and amounts at risk, e_A , are the adjusted values defined in Equation (4.3) which gave Figure 10. Figure 11 shows the fitted curves for ages 70 and 80 and both the parallel nature of the curves and the age dependent gaps are evident. Figure 12 shows the smoothed gaps, $B_a \hat{g}$, the gaps between the separate smooths for amounts and lives in Figure 10 averaged over years 1983–2000, and the gaps between $\log(R_A)$ and $\log(R_L)$

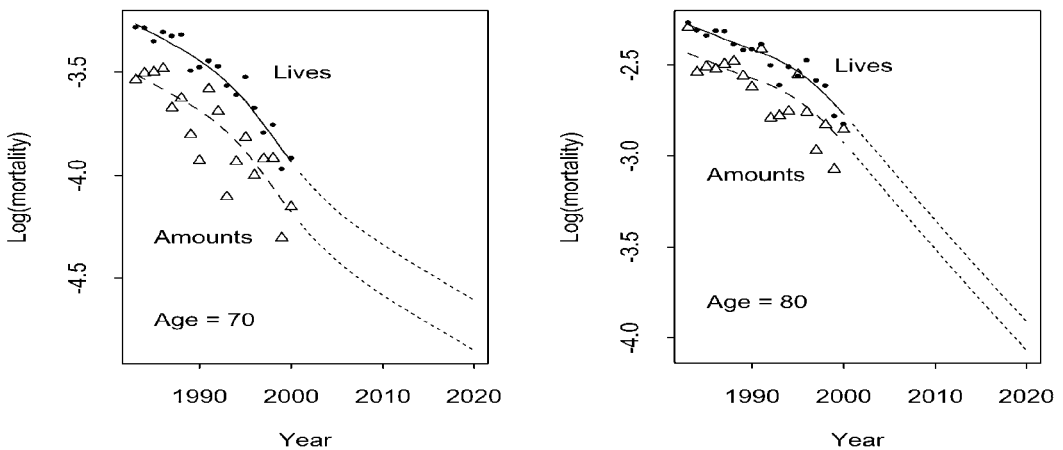


Figure 11 Observed, smoothed and extrapolated mortality (on the log scale) by lives and amounts with parallel cross sections by age: ages 70 and 80

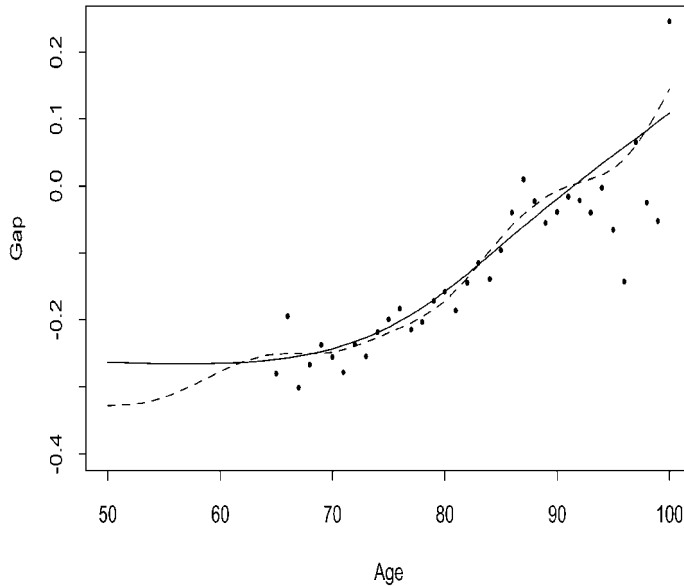


Figure 12 Smoothed gaps (—), gaps between separate smooths (---) and data gaps (points)

again averaged over years 1983–2000 [ages 50–64 are omitted because of missing values in $\log(R_A)$ and $\log(R_L)$].

We end this section by noting a simple connection between a joint model defined by Equation (4.4) and the idea of a ‘standard table’. The insurance industry makes much use of standard tables. A small number of such tables are used as reference tables and then tables for other classes of business are given in terms of age adjustments to the standard table. We illustrate the idea with the lives and amounts model. Suppose we take the lives mortality as the standard. Let $\hat{\eta}_L$ be the fitted linear predictor for lives. Now take $\eta_A = \hat{\eta}_L + \mathbf{1}_y \otimes B_a \mathbf{g}$, that is, $\hat{\eta}_L$ is an offset in the model with data $\mathbf{y}_A$ and $\mathbf{e}_A$, regression matrix $B = \mathbf{1}_y \otimes B_a$ and penalty $P = \lambda D_a' D_a$. The mortality by amounts is then summarized (with reference to the standard table) by $B_a \hat{\mathbf{g}}$ where $\hat{\mathbf{g}}$ is the vector of estimated regression coefficients.

5 Discussion

We have presented a flexible model for two-dimensional smoothing and shown how it may be fitted within the framework of a PGLM. We have concentrated on the smoothing of mortality tables for which the Poisson distribution is appropriate, but the method has much wider application and a number of generalizations. First, the method can be applied to any GLM with canonical link by replacing the weight matrix $W = \text{diag}(\boldsymbol{\mu})$ by a diagonal matrix with elements $w_{ii}^{-1} = (\partial \eta_i / \partial \mu_i)^2 \text{var}(y_i)$. Secondly, the method (which we applied to data on a regular grid) can be extended to deal with scattered data (Eilers *et al.*, 2004). Thirdly, computational issues, which are relatively minor for the data sets considered in this article, become a major consideration with

larger data sets. An important property of our model is that there exists an algorithm which reduces both storage requirements and computational times by an order of magnitude (Eilers *et al.*, 2004). Finally, our method of forecasting is suitable for the estimation of missing values or the extrapolation of data on a grid in any direction. All that is required are the regression and penalty matrices and the appropriate weight matrix. The extrapolation is then effected by Equation (3.20).

We emphasize the critical role of the order of the penalty, *pord*. The choice of the order of the penalty corresponds to a view of the future pattern of mortality: *pord* = 1, 2 or 3 correspond to future mortality continuing at a constant level, improving at a constant rate or improving at an accelerating (quadratic) rate, respectively. We do not believe *pord* = 1 is consistent with the data. A value of *pord* = 3 has some initial attractions; for example, in the right hand panel of Figure 6 *pord* = 3 gives an excellent fit to the future data. However, with a 50 year horizon, the quadratic extrapolation gives predictions which seem implausible and would certainly be viewed with alarm by the insurance industry.

The failure to predict accurately the fall in mortality rates has had far reaching consequences for the UK pensions and annuity business. What comfort can be drawn from the results presented in this article? We suggested in our introduction that any such predictions are unlikely to be correct. The difference between *P*-spline and Lee–Carter forecasts identified in Section 4.1 only serves to underline the difficulty of forecasting so far ahead. Our estimates of future mortality come with confidence intervals, and the widths of the confidence intervals indicate the level of uncertainty associated with the forecast. A prudent course is to allow for this uncertainty by discounting the predicted rates by a certain amount and our view is that some such discounting procedure is the only reasonable way of allowing for the uncertainty in these, or indeed any, predictions. The level of discount will have financial implications for the pricing and reserving of annuities and pensions.

Acknowledgements

We thank the CMIB for providing the data and for providing financial support to all three authors. We also thank the referees for their careful and constructive comments.

References

- Aerts M, Claeskens G and Wand MP (2002) Some theory for penalized spline generalized additive models. *Journal of Statistical Planning and Inference* **103**, 455–70.
- de Boor C (2001) *A practical guide to splines*. New York: Springer.
- Brouhns N, Denuit M and Vermunt JK (2002) A Poisson log-bilinear regression approach to the construction of projected lifetables. *Insurance: Mathematics & Economics* **31**, 373–93.
- Chatfield C (2003) Model selection, data mining and model uncertainty. *Proceedings of the 18th International Workshop on Statistical Modelling*. Leuven, Belgium: Katholieke Universiteit Leuven, 79–84.
- Clayton D and Schifflers E (1987) Models for temporal variation in cancer rates. II: Age-period-cohort models. *Statistics in Medicine* **6**, 469–81.
- Cleveland WS and Devlin SJ (1988) Locally weighted regression: an approach to regression

- analysis by local fitting. *Journal of the American Statistical Association* 83, 597–610.
- Coull BA, Ruppert D and Wand MP (2001a) Simple incorporation of interactions into additive models. *Biometrics* 57, 539–45.
- Coull BA, Schwartz J and Wand MP (2001b) Respiratory health and air pollution: additive mixed model analyses. *Biostatistics* 2, 337–50.
- Currie ID and Durban M (2002) Flexible smoothing with P -splines: a unified approach. *Statistical Modelling* 2, 333–49.
- Dierckx P (1993) *Curve and surface fitting with splines*. Oxford: Clarendon Press.
- Durban M and Currie ID (2003) A note on P -spline additive models with correlated errors. *Computational Statistics* 18, 251–62.
- Eilers PHC and Marx BD (1996) Flexible smoothing with B -splines and penalties. *Statistical Science* 11, 89–121.
- Eilers PHC and Marx BD (2002) Generalized linear additive smooth structures. *Journal of Computational and Graphical Statistics* 11, 758–83.
- Eilers PHC and Marx BD (2003) Multivariate calibration with temperature interaction using two-dimensional penalized signal regression. *Chemometrics and Intelligent Laboratory Systems* 66, 159–74.
- Eilers PHC, Currie ID and Durban M (2004) Fast and compact smoothing on large multidimensional grids. *Computational Statistics and Data Analysis*, in press.
- Gu C and Wahba G (1993) Semiparametric analysis of variance with tensor product thin plate splines. *Journal of the Royal Statistical Society: Series B* 55, 353–68.
- Hastie TJ and Tibshirani RJ (1990) *Generalized additive models*. London: Chapman & Hall.
- Hurvich CM, Simonoff JS and Tsai C-L (1998) Smoothing parameter selection in nonparametric regression using an improved Akaike information criterion. *Journal of the Royal Statistical Society: Series B* 60, 271–94.
- Lee TCM (2003) Smoothing parameter selection for smoothing splines: a simulation study. *Computational Statistics & Data Analysis* 42, 139–48.
- Lee RD and Carter LR (1992) Modeling and forecasting U.S. mortality. *Journal of the American Statistical Association* 87, 659–75.
- Lin X and Zhang D (1999) Inference in generalized additive mixed models by using smoothing splines. *Journal of the Royal Statistical Society: Series B* 61, 381–400.
- Marx BD and Eilers PHC (1998) Direct generalized additive modeling with penalized likelihood. *Computational Statistics & Data Analysis* 28, 193–209.
- Marx BD and Eilers PHC (1999) Generalized linear regression on sampled signals and curves: a P -spline approach. *Technometrics* 41, 1–13.
- Parise H, Wand MP, Ruppert D and Ryan L (2001) Incorporation of historical controls using semiparametric mixed models. *Journal of the Royal Statistical Society: Series C* 50, 31–42.
- Ruppert D (2002) Selecting the number of knots for penalized splines. *Journal of Computational and Graphical Statistics*, 11, 735–57.
- Ruppert D and Carroll RJ (2000) Spatially-adaptive penalties for spline fitting. *Australian and New Zealand Journal of Statistics* 42, 205–24.
- Schwarz G (1978) Estimating the dimension of a model. *Annals of Statistics* 6, 461–64.
- Searle SR (1982) *Matrix algebra useful for statistics*. New York: John Wiley & Sons.
- Wand MP (1999) On the optimal amount of smoothing in penalised spline regression. *Biometrika* 86, 936–40.
- Wand MP (2003) Smoothing and mixed models. *Computational Statistics* 18, 223–50.
- Wood SN (2003) Thin plate regression splines. *Journal of the Royal Statistical Society: Series B* 65, 95–114.
- Wood SN and Augustin NH (2002) GAMs with integrated model selection using penalized regression splines and applications to environmental modelling. *Ecological Modelling* 157, 157–77.